

§ 0.4

$$\begin{cases} \sin \theta \\ \cos \theta \\ \tan \theta \end{cases}$$

$$\begin{cases} \sec \theta = \frac{1}{\cos \theta} \\ \csc \theta = \frac{1}{\sin \theta} \\ \cot \theta = \frac{1}{\tan \theta} \end{cases}$$

$$\textcircled{1} \text{radian} = \text{度}(\text{degree}) \times \frac{\pi}{180^\circ}$$

$$\textcircled{2} \text{degree} = \text{radian} \times \frac{180^\circ}{\pi}$$

In exercises 1 and 2, convert the given radians measure to degrees.

1. ~~(a)~~ $\frac{\pi}{4}$ (b) $\frac{\pi}{3}$ (c) $\frac{\pi}{6}$ (d) $\frac{4\pi}{3}$

2. ~~(a)~~ $\frac{3\pi}{5}$ (b) $\frac{\pi}{7}$ (c) 2 (d) 3

In exercises 3 and 4, convert the given degrees measure to radians.

3. (a) 180° (b) 270° (c) 120° (d) 30°

~~4.~~ (a) 40° (b) 80° (c) 450° (d) 390°

1. (a) $\frac{\pi}{4} \times \frac{180^\circ}{\pi} = 45^\circ$

(b) $\frac{\pi}{3} \times \frac{180^\circ}{\pi} = 60^\circ$

2. (a) $\frac{3\pi}{5} \times \frac{180^\circ}{\pi} = 108^\circ$

(b) $\frac{\pi}{7} \times \frac{180^\circ}{\pi} = \frac{180^\circ}{7}$

4. (a) $40^\circ \times \frac{\pi}{180^\circ} = \frac{2}{9}\pi$

(b) $80^\circ \times \frac{\pi}{180^\circ} = \frac{4\pi}{9}$

(c) $450^\circ \times \frac{\pi}{180^\circ} = \frac{5\pi}{2}$

(d) $390^\circ \times \frac{\pi}{180^\circ} = \frac{13\pi}{6}$

In exercises 5–14, find all solutions of the given equation.

5. $2 \cos x - 1 = 0$

6. $2 \sin x + 1 = 0$

7. $\sqrt{2} \cos x - 1 = 0$

8. $2 \sin x - \sqrt{3} = 0$

9. $\sin^2 x - 4 \sin x + 3 = 0$

10. $\sin^2 x - 2 \sin x - 3 = 0$

11. $\sin^2 x + \cos x - 1 = 0$

12. $\sin 2x - \cos x = 0$

13. $\cos^2 x + \cos x = 0$

14. $\sin^2 x - \sin x = 0$

5. $2 \cos x - 1 = 0 \Rightarrow 2 \cos x = 1$

$$\Rightarrow \cos x = \frac{1}{2}$$

$$\Rightarrow x = \frac{\pi}{3} + 2k\pi, k \in \mathbb{Z} \text{ or } x = \frac{5\pi}{3} + 2k\pi, k \in \mathbb{Z}$$

9. $\sin^2 x - 4 \sin x + 3 = 0 \Rightarrow (\sin x - 1)(\sin x - 3) = 0$

$$\Rightarrow \sin x = 1 \text{ or } \sin x = 3$$

$$\Rightarrow x = \frac{\pi}{2} + 2k\pi, k \in \mathbb{Z}$$

10. $\sin^2 x - 2 \sin x - 3 = 0 \Rightarrow (\sin x - 3)(\sin x + 1) = 0$

$$\Rightarrow \sin x = -1$$

$$\Rightarrow x = \frac{3\pi}{2} + 2k\pi, k \in \mathbb{Z}$$

11. $\sin^2 x + \cos x - 1 \Rightarrow (1 - \cos^2 x) + \cos x - 1 = 0$

$$\Rightarrow \cos x(-\cos x + 1) = 0$$

$$\Rightarrow \cos x = 0 \text{ or } \cos x = 1$$

$$\Rightarrow x = \frac{\pi}{2} + k\pi, k \in \mathbb{Z} \text{ or } x = 2k\pi, k \in \mathbb{Z}$$

$$12. \sin 2x - \cos x = 0 \Rightarrow 2\sin x \cos x - \cos x = 0$$

$$\Rightarrow \cos x (2\sin x - 1) = 0$$

$$\Rightarrow \cos x = 0 \text{ or } \sin x = \frac{1}{2}$$

$$\Rightarrow x = \frac{\pi}{2} + k\pi, k \in \mathbb{Z} \text{ or}$$

$$x = \frac{\pi}{6} + 2k\pi, k \in \mathbb{Z} \text{ or}$$

$$x = \frac{5\pi}{6} + 2k\pi, k \in \mathbb{Z}$$

* Review + Push progress (Preview)

量 $\hat{v}$

$$1. a \sin x + b \cos x = \sqrt{a^2 + b^2} \left(\sin x \times \frac{\overset{\cos \theta}{a}}{\sqrt{a^2 + b^2}} + \cos x \times \frac{\overset{\sin \theta}{b}}{\sqrt{a^2 + b^2}} \right) \\ = \sqrt{a^2 + b^2} \sin(x + \theta), \text{ where } \begin{cases} \sin \theta = \frac{b}{\sqrt{a^2 + b^2}} \\ \cos \theta = \frac{a}{\sqrt{a^2 + b^2}} \end{cases}$$

$$2. a \cos x - b \sin x = \sqrt{a^2 + b^2} \left(\cos x \cdot \frac{a}{\sqrt{a^2 + b^2}} - \sin x \cdot \frac{b}{\sqrt{a^2 + b^2}} \right) \\ = \sqrt{a^2 + b^2} \cos(x + \theta), \text{ where } \begin{cases} \sin \theta = \frac{b}{\sqrt{a^2 + b^2}} \\ \cos \theta = \frac{a}{\sqrt{a^2 + b^2}} \end{cases}$$

§ 0.5

$f(x)$ 的 domain = x 所有可在 $f(x)$ 取值的 set

e.g. $f(x) = \frac{1}{x}$

Dom: $x \neq 0$ or $(-\infty, 0) \cup (0, \infty)$

e.g. $f(x) = \sqrt{x}$

Dom: $x \geq 0$ or $[0, \infty)$

52. A person who is 6 feet tall stands 4 feet from the base of a light pole and casts a 2-foot-long shadow. How tall is the light pole?
53. A surveyor stands 80 feet from the base of a building and measures an angle of 50° to the top of the steeple on top of the building. The surveyor figures that the center of the steeple lies 20 feet inside the front of the structure. Find the distance from the ground to the top of the steeple.

52. 一個人身高 6 英尺，站在距離燈柱底部 4 英尺處，並投射出長 2 英尺的影子。請問這根燈柱有多高？

53. 一名測量員站在距離建築物底部 80 英尺處，測得建築物頂部尖塔頂端的仰角為 50° 。測量員推算尖塔的中心位於該結構物正面往內 20 英尺處。求從地面到尖塔頂端的距離。

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交流电路

55. In an AC circuit, the voltage is given by $v(t) = v_p \sin(2\pi ft)$, where v_p is the peak voltage and f is the frequency in Hz. A voltmeter actually measures an average (called the **root-mean-square**) voltage, equal to $v_p / \sqrt{2}$. If the voltage has amplitude 170 and period $\pi/30$, find the frequency and meter voltage.

振幅

$$\textcircled{1} v(t) = V_p \sin(2\pi ft)$$

$$\textcircled{2} V_{\text{rms}} = \frac{V_p}{\sqrt{2}}$$

$$f = ?, V_{\text{rms}} = ?$$

